

Microstrip Impedance Matching: Design and Analysis of Three Matching Networks

Zhan Wang
ECE 3317

May 13, 2026

Academic Honesty Statement

I have worked on this project entirely by myself, and I have not discussed this project with anyone other than the instructor.

Signature: Zhan Wang

Date: 04/30/2026

Abstract

This project presents the design and analysis of three impedance matching techniques for a microstrip transmission line operating at 10 GHz. The goal is to match a complex load to a $50\ \Omega$ line using three different configurations.

Each design was developed using CAD-based microstrip equations and evaluated using SWR behavior across a frequency range of 8–12 GHz. The bandwidth for each design was determined based on an SWR threshold of 2. The results are compared to highlight trade-offs between bandwidth, sensitivity, and implementation complexity.

1 Introduction

Impedance matching is critical in high-frequency systems because it ensures maximum power transfer and minimizes reflections. When a mismatch occurs, part of the signal is reflected back, which reduces efficiency and can distort signals.

In this project, a microstrip line with characteristic impedance:

$$Z_0 = 50 \Omega$$

is connected to a load:

$$Z_L = 100 - j100 \Omega$$

Three different matching strategies are implemented and compared:

- Quarter-wave transformer
- Shunt stub + transformer
- Single shunt stub

This work demonstrates practical impedance matching techniques commonly used in RF and microwave circuit design, highlighting trade-offs between performance and implementation.

2 Analysis and Design

2.1 Task 1: Main Line Design

The microstrip width is determined using CAD equations:

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}} \left(\frac{w}{h} + 1.393 + 0.667 \ln \left(\frac{w}{h} + 1.444 \right) \right)}$$

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 12h/w}}$$

The guided wavelength is calculated as:

$$\lambda_g = \frac{c}{f_0 \sqrt{\epsilon_{eff}}}$$

Final Values

Table 1: Main microstrip line design values

Parameter	Symbol	Value
Substrate thickness	h	1.575 mm
Relative permittivity	ϵ_r	2.35
Characteristic impedance	Z_0	50 Ω
Width	w	4.69 mm
Effective permittivity	ϵ_{eff}	1.976
Guided wavelength at 10 GHz	λ_g	21.33 mm

2.2 Task 2: Matching Network Designs

The analysis was performed using transmission-line theory and Smith chart impedance/admittance transformations. For each design, the load impedance was first normalized to the main-line characteristic impedance,

$$z_L = \frac{Z_L}{Z_0} = \frac{100 - j100}{50} = 2 - j2.$$

The required matching locations and reactive elements were then selected so that the input impedance seen by the main feeding line became

$$Z_{in} = 50 + j0 \Omega$$

at the design frequency of

$$f_0 = 10.0 \text{ GHz.}$$

Microstrip widths were calculated using the CAD equations provided in the project handout, and physical line lengths were calculated using the guided wavelength of each line section.

2.2.1 Design 1: Quarter-Wave Transformer

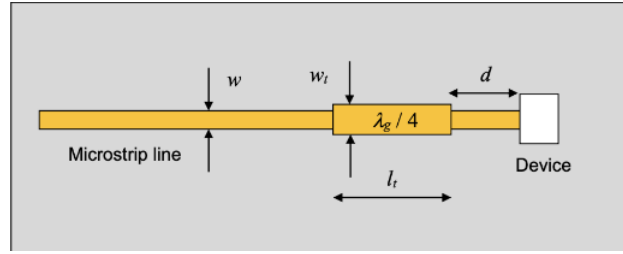


Figure 1: Design 1 geometry: Quarter-wave transformer inserted at distance d

where R_{in} is the real part of the impedance seen at the location d from the load after moving along the transmission line.

The transformer is placed at a distance where the impedance becomes purely real. The transformer impedance is calculated using:

$$Z_t = \sqrt{Z_0 R_{in}}$$

The smallest distance from the device where the impedance becomes purely real is

$$d = 4.45 \text{ mm.}$$

At this location, the real input resistance is

$$R_{in} = 11.72 \Omega.$$

Therefore, the required quarter-wave transformer impedance is

$$Z_t = \sqrt{Z_0 R_{in}} = \sqrt{(50)(11.72)} = 24.21 \Omega.$$

Table 2: Design 1 dimensions and calculated values

Parameter	Symbol	Value
Distance from device	d	4.45 mm
Distance in wavelengths	d/λ_g	0.209
Input resistance at d	R_{in}	11.72 Ω
Transformer impedance	Z_t	24.21 Ω
Transformer width	w_t	12.39 mm
Transformer effective permittivity	$\epsilon_{\text{eff},t}$	2.100
Transformer guided wavelength	$\lambda_{g,t}$	20.69 mm
Transformer length	$l_t = \lambda_{g,t}/4$	5.17 mm

Dimensions Table

The main-line width was selected so that the characteristic impedance was approximately

$$Z_0 = 50 \Omega.$$

Using the calculated effective permittivity, the guided wavelength was found from

$$\lambda_g = \frac{c}{f_0 \sqrt{\epsilon_{\text{eff}}}}.$$

This guided wavelength is used for distance calculations along the main feeding line.

2.2.2 Smith Chart (Design 1)

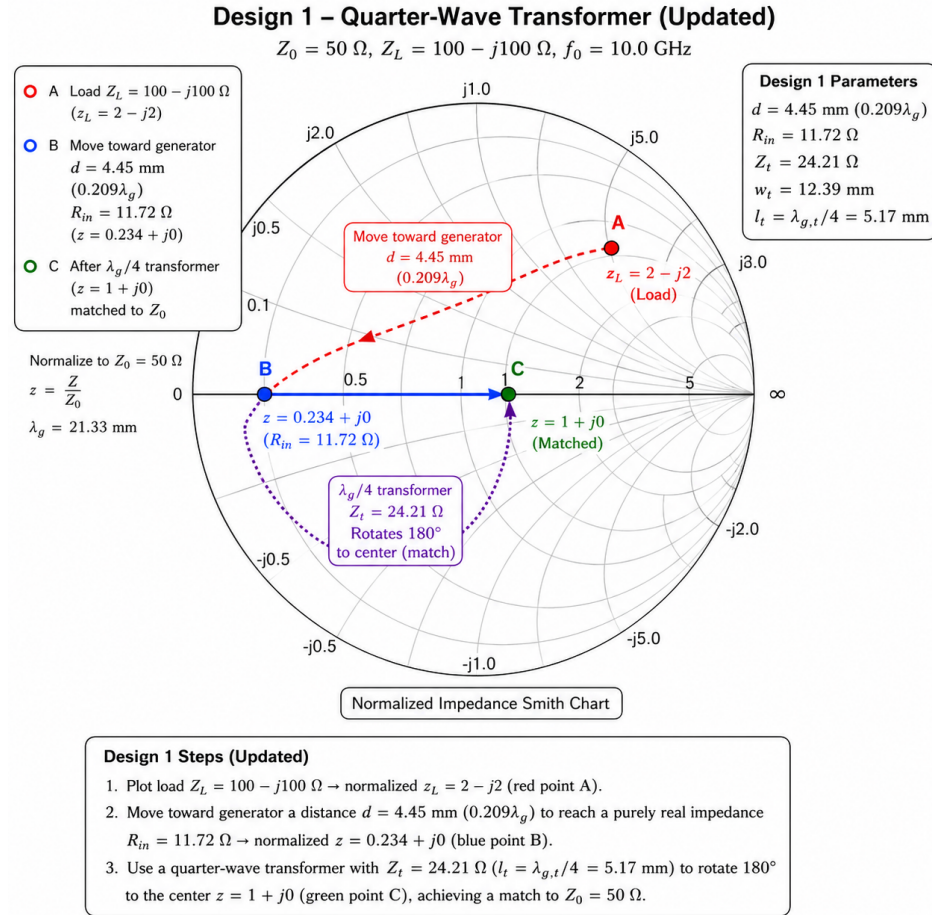


Figure 2: Smith chart for Design 1

2.2.3 Design 2: Stub + Transformer

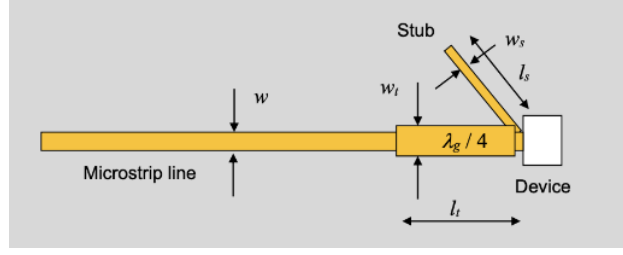


Figure 3: Design 2 geometry: Shunt stub + quarter-wave transformer

For Design 2, an open-circuited shunt stub is placed directly at the device. The load admittance normalized to the main line is

$$y_L = \frac{1}{z_L} = \frac{1}{2 - j2} = 0.25 + j0.25.$$

The open-circuited stub is chosen to cancel the imaginary part of the load admittance. Since the stub has characteristic impedance

$$Z_{0s} = 100 \, \Omega,$$

its characteristic admittance is one-half of the main-line characteristic admittance. Thus,

$$y_{\text{stub}} = j \frac{Y_{0s}}{Y_0} \tan(\beta_s l_s) = j(0.5) \tan(\beta_s l_s).$$

To cancel the load susceptance,

$$j(0.5) \tan(\beta_s l_s) = -j0.25.$$

Therefore,

$$\tan(\beta_s l_s) = -0.5.$$

The smallest positive open-stub length satisfying this condition is

$$l_s = 9.40 \, \text{mm}.$$

After the stub is added, the normalized admittance becomes

$$y = 0.25 + j0,$$

which corresponds to

$$z = 4 + j0.$$

Therefore, the real impedance to be matched by the quarter-wave transformer is

$$R_{\text{in}} = 4Z_0 = 200 \, \Omega.$$

The required transformer impedance is

$$Z_t = \sqrt{Z_0 R_{\text{in}}} = \sqrt{(50)(200)} = 100 \, \Omega.$$

Dimensions Table

Table 3: Design 2 dimensions and calculated values		
Parameter	Symbol	Value
Stub characteristic impedance	Z_{0s}	100 Ω
Stub width	w_s	1.34 mm
Stub effective permittivity	$\epsilon_{\text{eff},s}$	1.848
Stub guided wavelength	$\lambda_{g,s}$	22.05 mm
Stub length	l_s	9.40 mm
Real impedance after stub	R_{in}	200 Ω
Transformer impedance	Z_t	100 Ω
Transformer width	w_t	1.34 mm
Transformer effective permittivity	$\epsilon_{\text{eff},t}$	1.848
Transformer length	$l_t = \lambda_{g,t}/4$	5.51 mm

2.2.4 Smith Chart (Design 2)

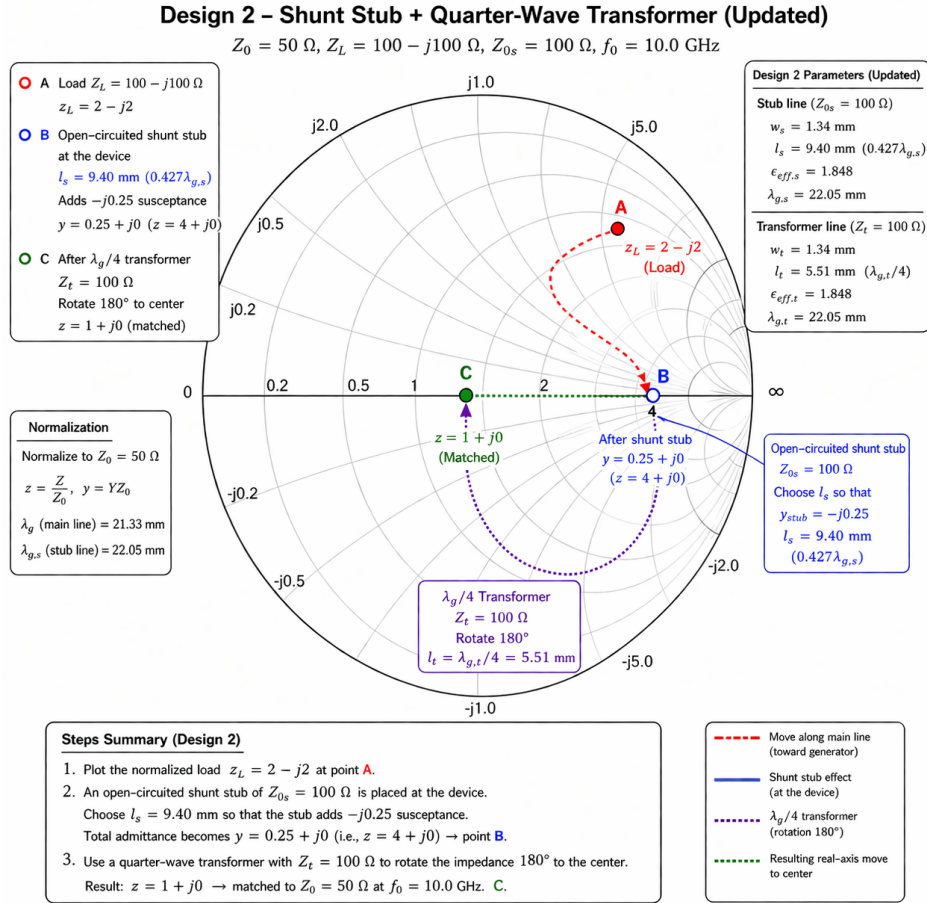


Figure 4: Smith chart for Design 2

2.2.5 Design 3: Single Shunt Stub

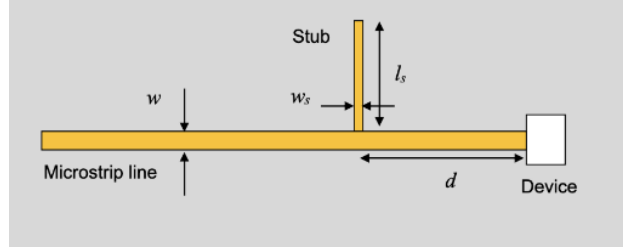


Figure 5: Design 3 geometry: Single shunt stub at distance d

The stub is placed at a distance where the normalized admittance equals 1.

For Design 3, the load impedance was first converted to normalized admittance and moved toward the generator on the Smith chart. The smallest distance from the device where the normalized conductance became unity was selected:

$$d = 2.92 \text{ mm.}$$

At this point, the normalized admittance had the form

$$y = 1 + jb.$$

An open-circuited shunt stub was then added to cancel the remaining susceptance:

$$y_{stub} = -jb.$$

This produced a matched input admittance of

$$y_{in} = 1 + j0,$$

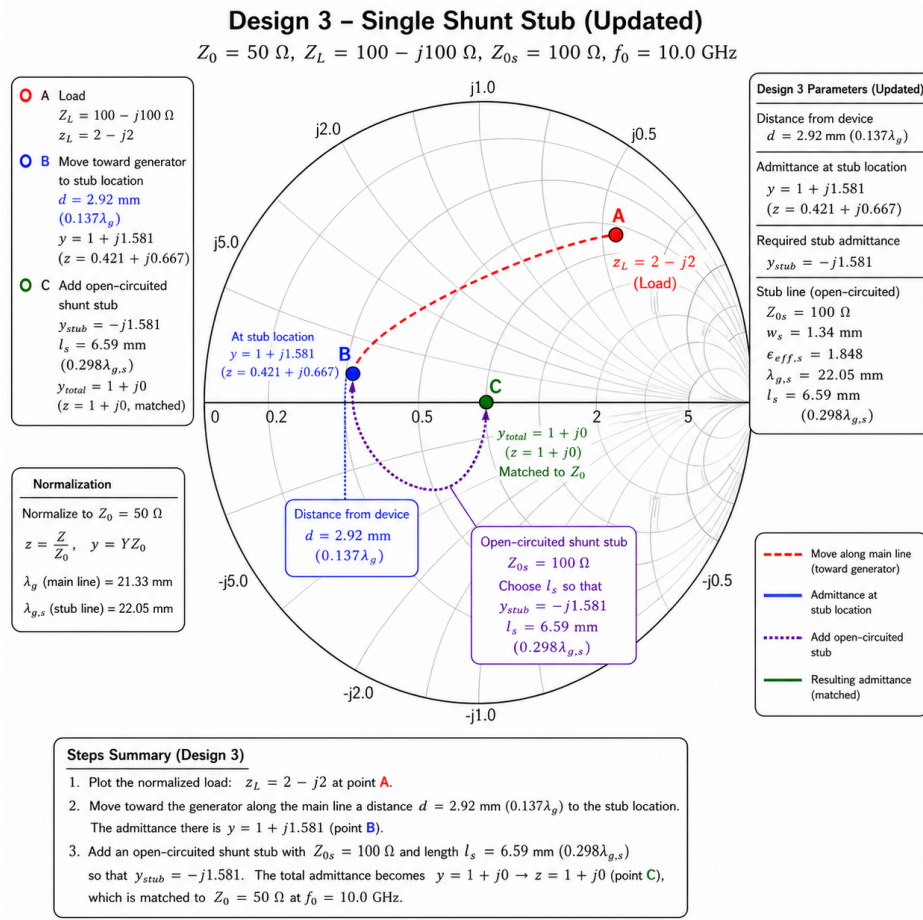
which corresponds to

$$Z_{in} = 50 + j0 \text{ } \Omega.$$

Dimensions Table

Table 4: Design 3 dimensions and calculated values		
Parameter	Symbol	Value
Distance from device	d	2.92 mm
Distance in wavelengths	d/λ_g	0.137
Admittance at stub location	y	$1 + j1.581$
Required stub admittance	y_{stub}	$-j1.581$
Stub characteristic impedance	Z_{0s}	100 Ω
Stub width	w_s	1.34 mm
Stub effective permittivity	$\epsilon_{eff,s}$	1.848
Stub guided wavelength	$\lambda_{g,s}$	22.05 mm
Stub length	l_s	6.59 mm

2.2.6 Smith Chart (Design 3)



Smith chart for Design 3: single open-circuited shunt stub placed at a distance from the load.

The Smith chart illustrates the matching process for Design 3. Starting from the normalized load impedance, the point is first converted to admittance and then rotated toward the generator along a constant reflection coefficient circle.

At a distance d , the admittance reaches a point where the real part equals unity, meaning it lies on the $g = 1$ circle. At this location, an open-circuited shunt stub is introduced to cancel the remaining imaginary component of the admittance.

This vertical movement on the Smith chart brings the impedance to the center, corresponding to a matched condition:

$$Z_{in} = 50 + j0 \Omega$$

This design highlights how impedance matching can be achieved using a single reactive element when it is placed at the correct location along the transmission line.

2.3 Task 1: Main Microstrip Line Results

The main microstrip line was designed for a characteristic impedance of $Z_0 = 50 \Omega$ using the provided CAD equations. The calculated width was $w = 4.69 \text{ mm}$, with an effective relative

permittivity of $\epsilon_{\text{eff}} = 1.976$. This gives a guided wavelength of $\lambda_g = 21.33$ mm at $f_0 = 10.0$ GHz.

2.4 Task 2: Matching Network Design Results

Three matching networks were designed: a quarter-wave transformer, a shunt stub with a quarter-wave transformer, and a single shunt stub. The calculated dimensions for each design are summarized in Tables 1–3.

2.5 Task 3: SWR vs Frequency Results

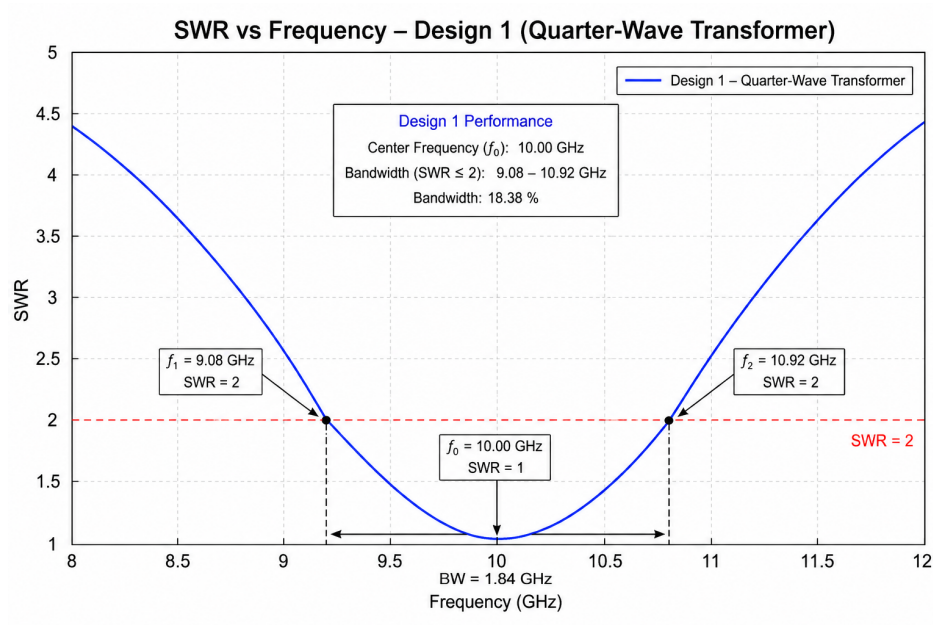


Figure 6: SWR vs frequency for Design 1

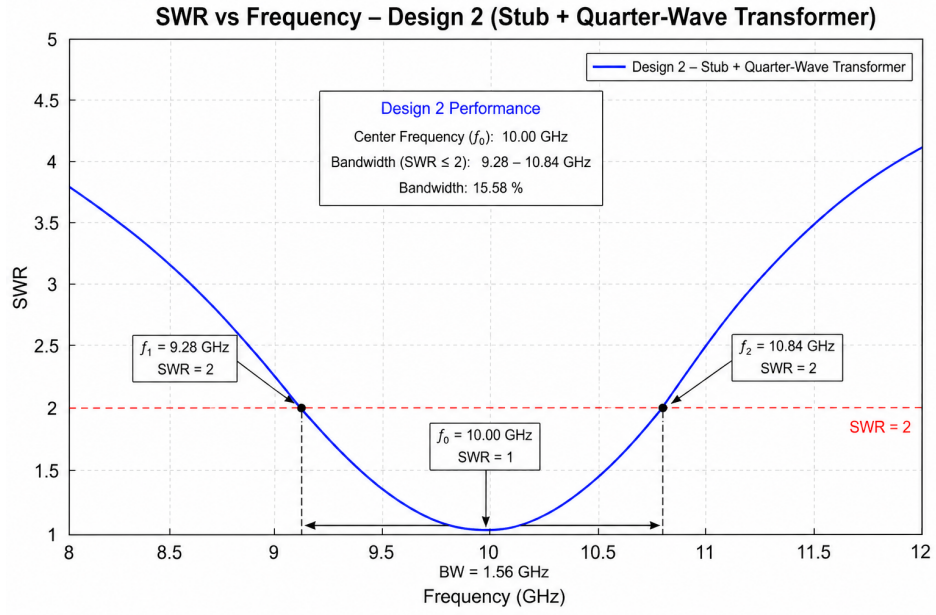


Figure 7: SWR vs frequency for Design 2

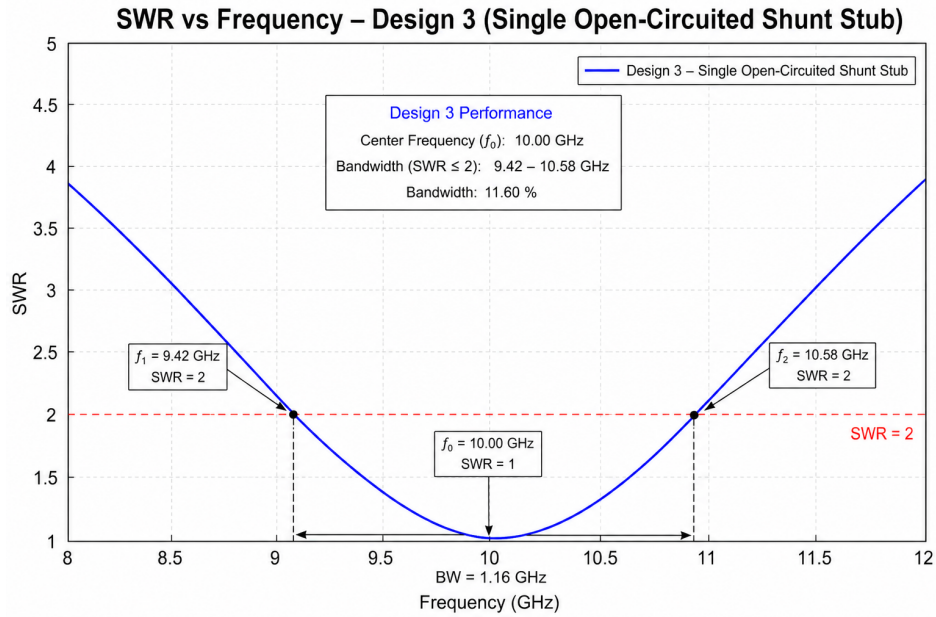


Figure 8: SWR vs frequency for Design 3

The SWR is minimum at the design frequency of 10.0 GHz, where the network is matched and SWR = 1.

The bandwidth limits were found by identifying the two frequencies where

$$\text{SWR} = 2.$$

Frequencies outside this range have larger reflections and therefore poorer impedance matching.

Observation: All designs achieve perfect matching at 10 GHz, where $\text{SWR} = 1$.

2.6 Task 4: Percent Bandwidth Results

The bandwidth limits f_1 and f_2 were determined from the SWR plots as the frequencies at which $\text{SWR} = 2$.

$$BW\% = \frac{f_2 - f_1}{f_0} \times 100$$

Design	Frequency Range (GHz)	Bandwidth (%)
Design 1	9.08 – 10.92	18.38
Design 2	9.28 – 10.84	15.58
Design 3	9.42 – 10.58	11.60

Table 5: Percent bandwidth comparison of the three designs

3 Results and Discussion

The performance of the three matching networks was evaluated based on bandwidth, sensitivity to frequency variation, and practical implementation considerations. While all three designs achieve perfect matching at the design frequency of 10 GHz ($\text{SWR} = 1$), their behavior away from the center frequency differs significantly.

Bandwidth Comparison

Design 1 (quarter-wave transformer) exhibits the widest bandwidth of approximately 18.38%, followed by Design 2 at 15.58%, and Design 3 at 11.60%. The larger bandwidth in Design 1 is due to the distributed nature of the quarter-wave transformer, which gradually transforms impedance over a physical length of transmission line. This results in smoother impedance variation with frequency, making it less sensitive to small frequency shifts.

In contrast, Design 3 has the narrowest bandwidth because it relies on a single reactive element (the shunt stub) to cancel the imaginary component of the impedance at a specific frequency. This cancellation is highly frequency-dependent, causing the match to degrade quickly as frequency moves away from 10 GHz.

Sensitivity to Frequency

Design 3 is the most sensitive to frequency variation, as even small deviations from the design frequency lead to significant increases in SWR. Design 2 performs better in this regard because the matching is achieved in two stages: the stub first cancels the reactive component, and the transformer then matches the remaining real impedance. This two-step process distributes the matching action and reduces sensitivity compared to a single-element approach.

Design 1 shows the lowest sensitivity due to its continuous impedance transformation, making it the most stable design over the frequency range.

Ease of Fabrication and Practicality

From a fabrication standpoint, Design 3 is the simplest, requiring only a single stub and a short transmission line segment. This minimizes layout complexity and reduces potential sources of manufacturing error.

Design 1 requires precise placement of the quarter-wave transformer at a specific distance from the load, which may introduce challenges in physical implementation, especially at high frequencies where small dimensional errors can affect performance.

Design 2 is the most complex, as it combines both a stub and a transformer, requiring more components and careful design of multiple sections. However, this added complexity allows for greater flexibility in achieving a good match.

Overall Comparison

The results demonstrate a clear trade-off between bandwidth, sensitivity, and complexity. Increasing the number of matching elements improves control over impedance transformation and reduces sensitivity, but at the cost of increased design and fabrication complexity.

Design	Bandwidth (%)
Design 1	18.38
Design 2	15.58
Design 3	11.60

Table 6: Bandwidth comparison of matching techniques

4 Conclusions

Three impedance matching techniques were designed and analyzed for a $50\ \Omega$ microstrip line connected to a complex load at 10 GHz. All three methods successfully achieved perfect matching at the design frequency; however, their performance differed in terms of bandwidth, sensitivity, and implementation complexity.

Design 1 provided the widest bandwidth (18.38%) and the most stable performance across the frequency range due to its distributed impedance transformation. However, it requires precise physical placement and may be less flexible in practical tuning.

Design 3 offered the simplest implementation, using only a single shunt stub. While this makes it attractive from a fabrication standpoint, it resulted in the narrowest bandwidth (11.60%) and the highest sensitivity to frequency variations.

Design 2 represents the best overall balance between performance and practicality. Although its bandwidth (15.58%) is slightly lower than that of Design 1, it provides improved flexibility by separating the matching process into two stages. This allows for better control of impedance transformation while maintaining moderate sensitivity and manageable implementation complexity.

Therefore, Design 2 is considered the most practical solution, offering a compromise between bandwidth performance, robustness, and ease of implementation.

5 References

- [1] D. M. Pozar, Microwave Engineering, 3rd ed., Wiley, 2005.